

Food Inflation Forecasting - Model Comparison Report

1. Machine Learning Models

Three forecasting approaches were evaluated: - Linear Regression baseline - Random Forest Regressor - Gradient Boosting Regressor Models were trained using historical monthly observations.

2. Evaluation Metrics

Performance was measured using: MAE: Average prediction error. RMSE: Penalizes larger prediction errors. MAPE: Measures percentage prediction error.

3. Model Selection

The final selected model was Gradient Boosting. Evaluation results: Model MAE RMSE MAPE
Random Forest 17.949067 22.193873 13.252057 Gradient Boosting 16.235935 20.699927 11.906682